

Communication over power lines

Delhi Technological University, is working at using the existing power infrastructure for communication. It has tested a deployment at Alipur that can transmit information over the power lines, including the internet and multimedia.

Abracadabra!

Although not a new concept, wireless power is now being implemented. We look at some of the more interesting developments in the leading labs working on wireless power

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The wired world

It has taken about a century of digging and cabling, to network the world together. Electricity grids around the world use wires to take electricity from where it is generated to deliver it to where electricity is needed. These setups typically involve the massive plants, transmission towers, and the number of pylons on the distribution grid. When electricity was being introduced, wiring up so many places, across the world, seemed like a daunting task. Wires, were however relatively cheap, as well as efficient, and so wired electricity became the de facto standard of distributing electricity.

Wireless power in itself is not a new idea, it had been conceptualised more than a century ago, only never really implemented. In 1893, the Chicago World Fair brought together trends from around the world. It was also one of the earliest large scale demonstrations on the uses of electricity. Nikola Tesla won a bid to arrange for electricity for the entire fair (Thomas Edison and JP Morgan were his competitors). Here, the attendees saw the earliest public demonstration of wireless electricity transmission, a setup by Tesla. Tesla's approach used something known as electrostatic induction. The method involves suspending two sheets of metal from the ceiling, one on either side of the room, and creating an alternating electrostatic field between them. This allows light bulbs to light up anywhere within a room.

Tesla's vision for wireless power was to juice up devices anywhere in an environment, without the devices being

tethered to wires. Even in 1893, the fact that bulbs were tethered to wires, making them immobile, was considered a problem. Over the years, many successful methods to deliver energy over distances wirelessly have been developed. Fundamentally, wireless transfer of information was considered similar to the wireless transfer of energy. Applications for these have been many. These included remote control devices, vehicles in a closed environment and communication.

About two decades ago, a range of electronic devices and household appliances started showing up in the market. These ranged from digital cameras and spare television sets to vacuum cleaners. All of these needed plug points, and the number of points often fell short. A typical user would face a number of problems, with just the wires. Sometimes the plugs would be of a different standard than the socket, almost every device had a unique adaptor that was not compatible with anything else, and running a number of devices at the same time became difficult. In an environment where the density of power-hungry devices is large, the wiring invariably became a tangled mess that nobody could navigate. A cramped office with computers lined up on a table, is a perfect example of this.

While wireless power over large distances is not going to be feasible any time soon, there are two areas where users stand to benefit a lot from wireless power. Wireless power is great for sending a large amount of power over small distances, as in charging all

devices in a single room, or household, and sending a small amount of power over large areas, say, all the walkie-talkies on a construction site. A company called Splashpower toyed around with early prototypes of wireless energy transfer as early as 2001. Demonstrations of technologies have been around since 2008, showing up at conferences such as CES, and are expected to be implemented by 2011.

Near range wireless power

The technologies for existing near-range wireless electricity transmission are all based on the principle of induction.

Induction is a phenomenon that causes electricity to be created when there is a change in a magnetic field. There are many kinds of inductive phenomena, including electromagnetic and electrostatic induction. Wireless transfer of a small amount of energy to and from devices using these methods, is pretty common already. The tags in retail outlets, for example, use this technology, as do some smartcards and the devices that detect shoplifting. In some highly-specific applications of electricity, which can be potentially harmful, some of these technologies are used to power devices that do not require much electricity. Examples of such applications range from electric cookers to electric toothbrushes.

Companies such as WiPower are trying to up the ante a bit, by allowing consumer electronics such as cameras, laptops, PMPs, mobile phones and even televisions to charge up wirelessly. WiPower uses a form of electromagnetic induction known as "resonant cou-

